

Activity: cumulant generating function

Cumulant generating function

Let Y be a random variable. The **cumulant generating function** (CGF) of Y is the log of the moment generating function:

$$C(t) = \log \mathbb{E}[e^{tY}]$$

Show that

$$\left. \frac{d}{dt} C(t) \right|_{t=0} = \mathbb{E}[Y] \qquad \left. \frac{d^2}{dt^2} C(t) \right|_{t=0} = \text{Var}(Y)$$